

Propeller Bench Test Analysis Report

MT_2216_10x4.5

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Step 1 — Polynomial fits (throttle → RPM / Thrust / Torque)

Channel	Degree	R²
rpm	1	0.99762
thrust	2	0.99832
torque	3	0.99805

Equations:

$rpm(x) = 70.3939 \cdot x + 2991.67$
 $thrust(x) = 0.000679454 \cdot x^2 + 0.0679155 \cdot x + 1.00322$
 $torque(x) = -1.23061e-07 \cdot x^3 + 2.73244e-05 \cdot x^2 + 0.000448811 \cdot x + 0.0225743$

Step 2 — Lag lookup tables (see lag_vs_throttle_rate and lag_vs_rpm CSVs)

Step 3 — Lambda model: $RPM_{rate} = \lambda \cdot (RPM_{cmd} - RPM)$ | variant comparison

Variant	N	R2	Rmse	Nrmse	Λ Mean	Λ Std	Λ Min	Λ Max	Frac Negative
direct_all	9837	0.1980	5.1195	1.252	4.088	5.717	-18.983	28.216	0.313
direct_filtered	8842	0.3220	4.7884	1.334	3.590	5.815	-19.147	28.591	0.349
lag_all	12118	0.5113	2.6750	0.187	14.322	3.826	8.167	18.379	0.000
lag_filtered	11118	0.5262	2.6366	0.182	14.503	3.830	8.167	18.379	0.000

Methodology & Output Guide

Step 3 — Lambda Estimation Approaches

The first-order RPM model is: $\text{RPM_rate} = \lambda \cdot (\text{RPM_commanded} - \text{RPM})$

λ (lambda, s⁻¹) is the bandwidth coefficient. Two independent methods estimate it:

- DIRECT** Rearranges the model equation at every time step:
- $\lambda = \text{RPM_rate} / (\text{RPM_cmd} - \text{RPM})$
 - RPM_rate = Savitzky-Golay numerical derivative of measured RPM.
 - RPM_cmd = global polynomial fit (Step 1) evaluated at current throttle.
 - Requires $|\text{RPM_cmd} - \text{RPM}| \geq 50$ RPM to avoid noisy near-steady-state estimates.
 - Reflects the instantaneous, model-equation-based lambda at each sample.
- LAG** Inverts the empirical time lag between throttle input and RPM response:
- $\lambda = 1 / \tau = 1000 / \text{lag_rpm_ms}$
 - lag_rpm_ms is measured in Step 2 via threshold-crossing analysis – the time for RPM to cross 30/40/50/60/70 % of its step range after the corresponding throttle crossing. Treating this lag as the first-order time constant τ gives $\lambda = 1/\tau$. This is a system-identification estimate derived entirely from the measured response timing, independent of the model equation.

Lambda Fit Metrics Interpretation

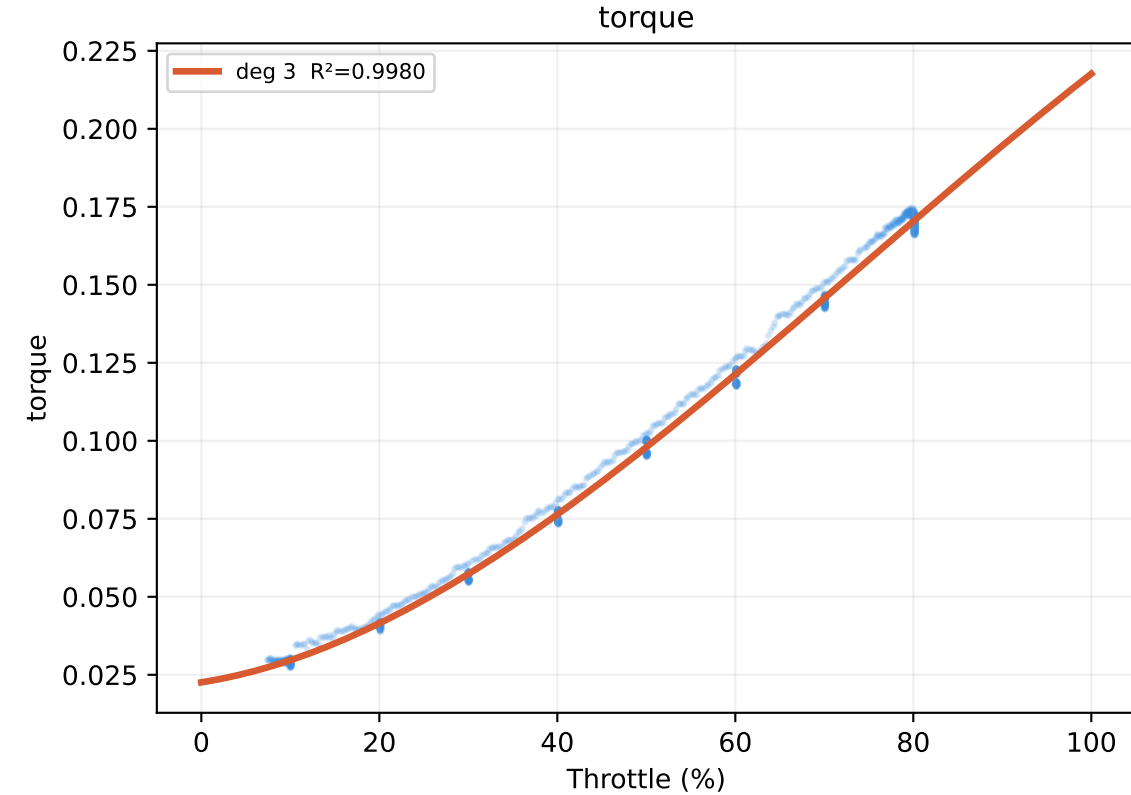
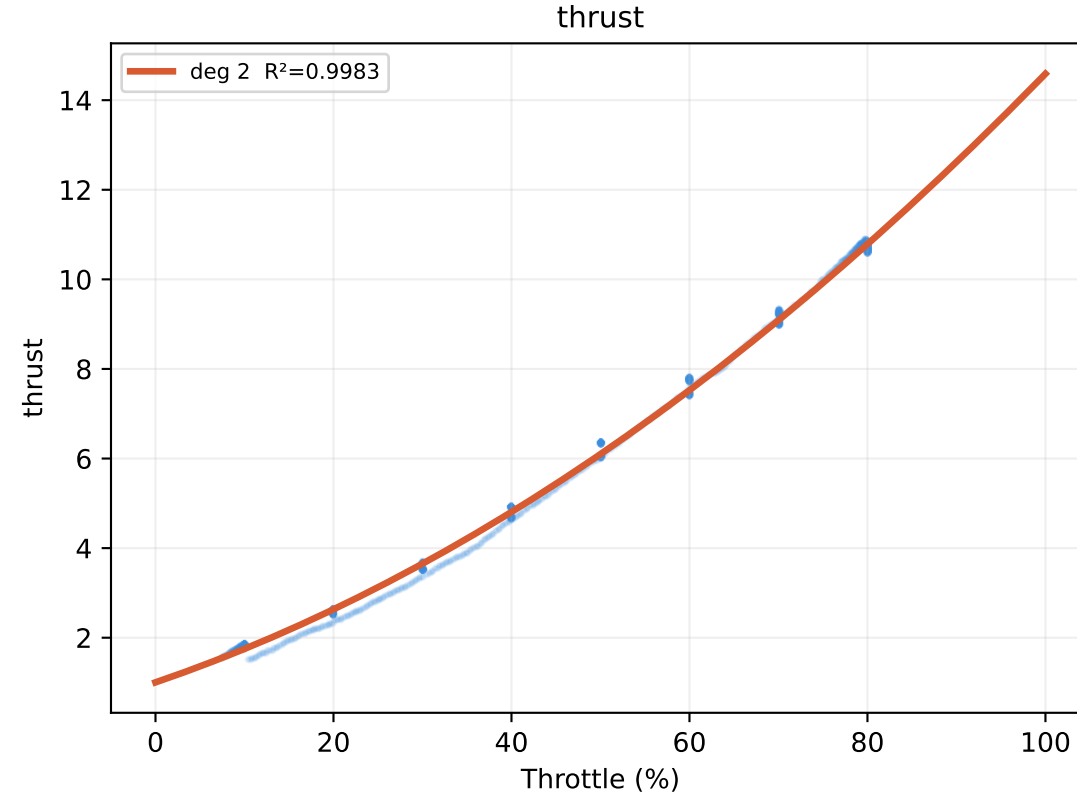
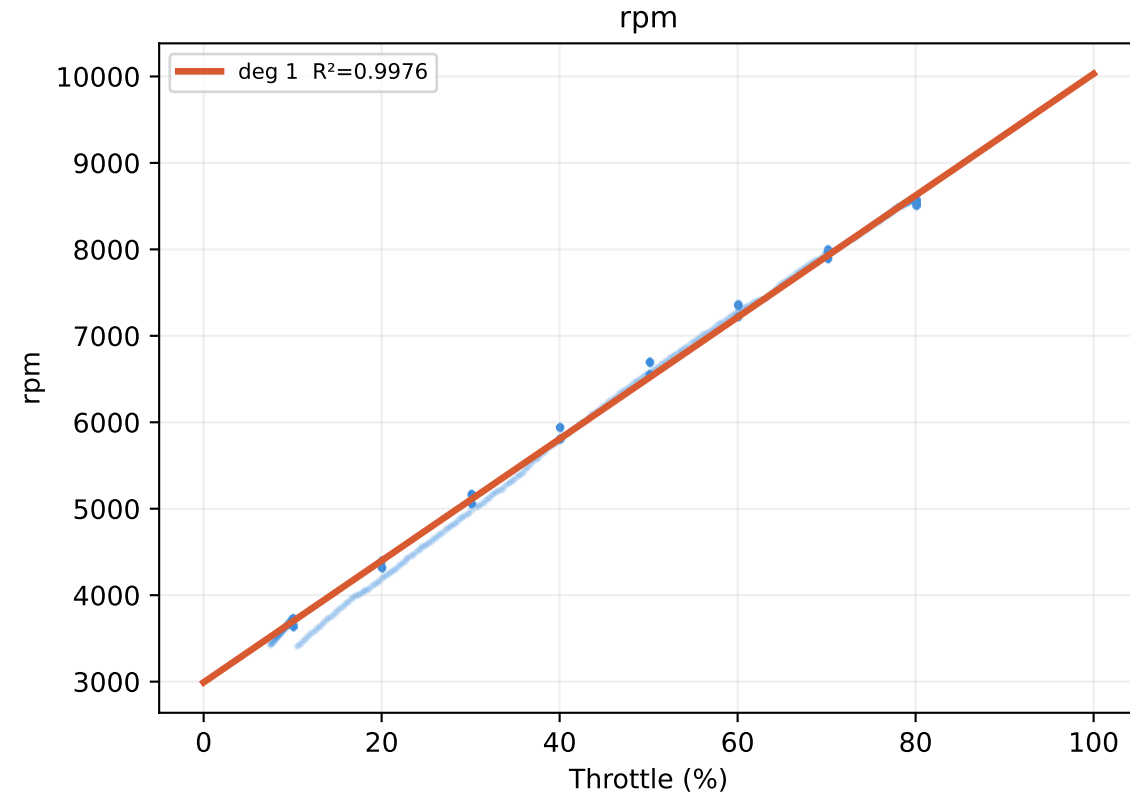
- R²** Explained variance of the bivariate polynomial $\lambda(\text{RPM}, \text{RPM_rate})$. Higher is better. < 0.3 means λ has little dependence on RPM / rate.
- RMSE** Root-mean-square prediction error (s⁻¹). Lower is better.
- NRMSE** RMSE / |mean λ |. Scale-free; < 0.5 indicates a well-conditioned fit.
- λ mean** Physically expect ~5–15 s⁻¹ (= 1/lag; lag ≈ 70–200 ms).
- λ std** Spread of observed λ values across the flight envelope.
- λ min/max** Check for physical plausibility – all values should be positive.
- neg %** Fraction of $\lambda < 0$ (unphysical); 0 % is ideal.
- filter Δ** Rows dropped by the $|\text{RPM_rate}|$ threshold. 0 means threshold too high.
- |rate|max** Largest $|\text{RPM_rate}|$ retained; compare to threshold to assess filter.

Output Files

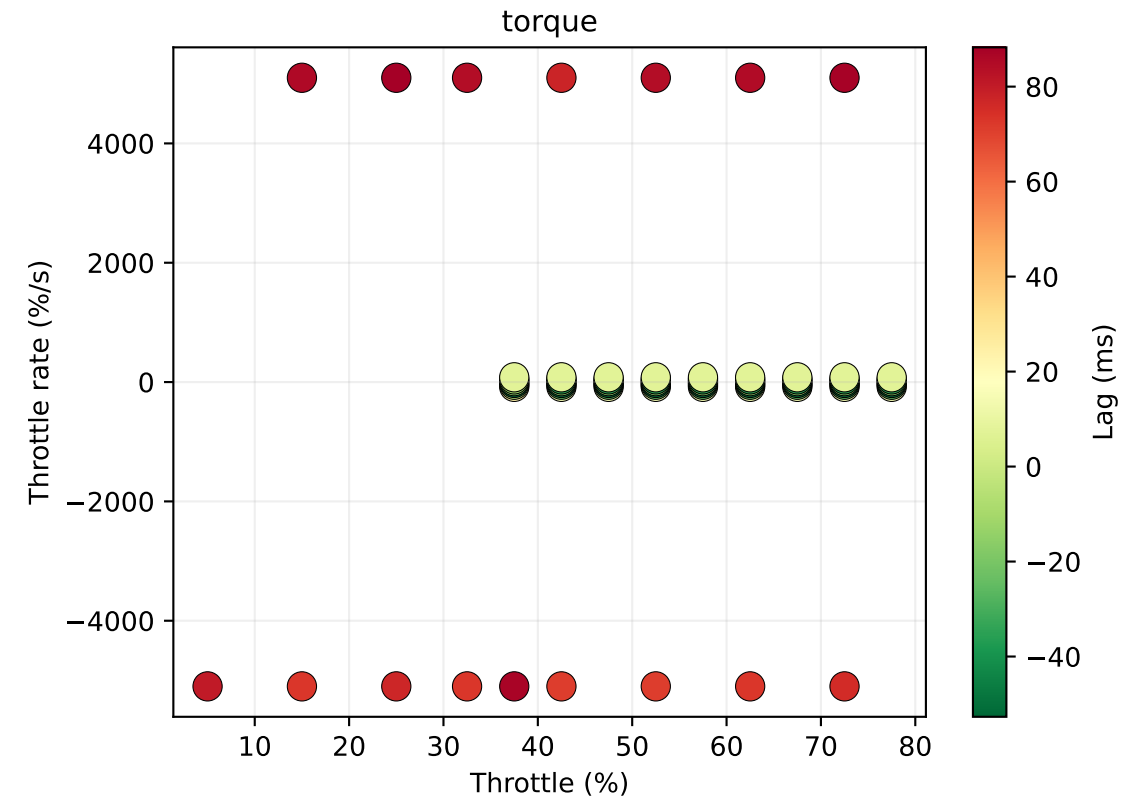
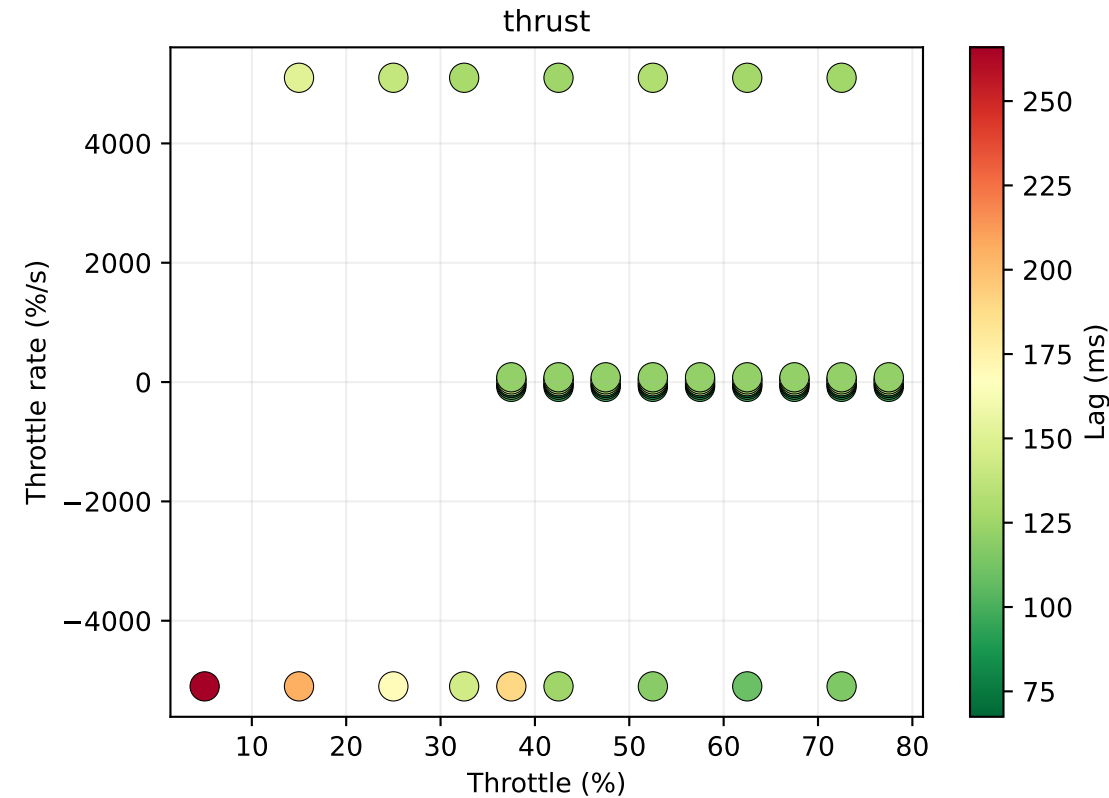
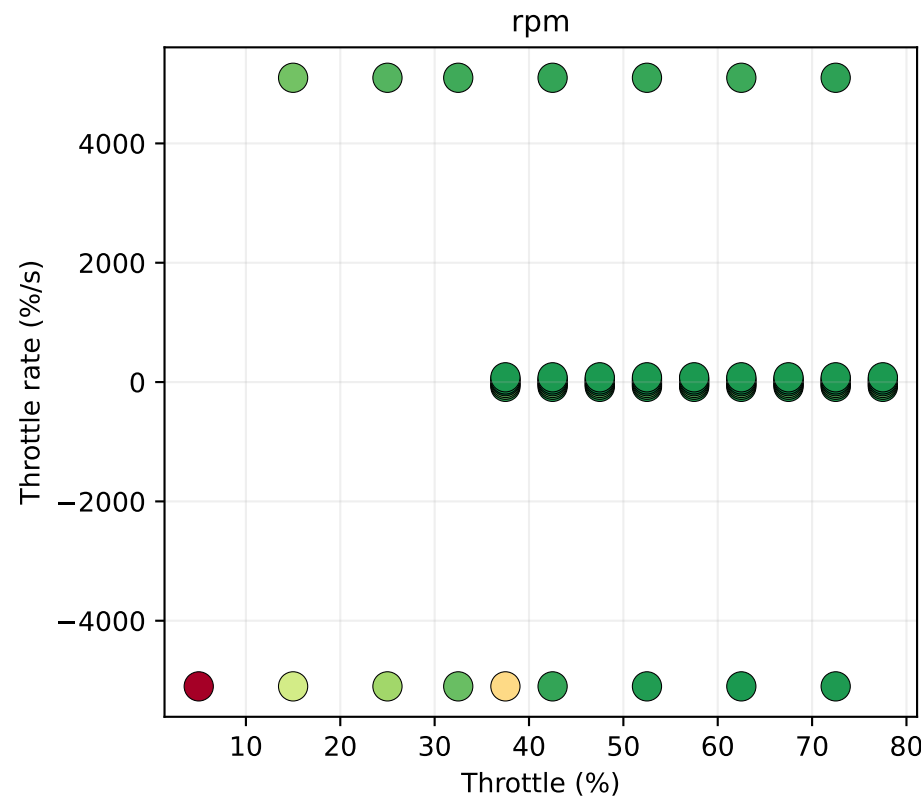
- global_fit_coeffs.csv** Polynomial coefficients: throttle → RPM / Thrust / Torque
- global_fit_results.txt** Human-readable fit report with equations and R²
- lag_vs_throttle_rate.csv** Weighted mean lag (ms) binned by throttle % × throttle rate %/s
- lag_vs_rpm.csv** Thrust & torque lag (ms) binned by mean RPM
- lambda_lookup_<variant>.csv** Binned λ table (RPM × RPM_rate) for each variant
- lambda_fit_coeffs_<variant>.csv** Bivariate polynomial coefficients for $\lambda(\text{RPM}, \text{RPM_rate})$
- lambda_comparison.csv** Side-by-side metrics for all four variants
- <motor>_report.pdf** This report (all figures + title + methodology pages)

- Variants: **direct_all** / **direct_filtered** – direct approach, unfiltered / rate-limited
 lag_all / **lag_filtered** – lag-based approach, unfiltered / rate-limited

Step 1 — Global weighted polynomial fits (pooled step data)

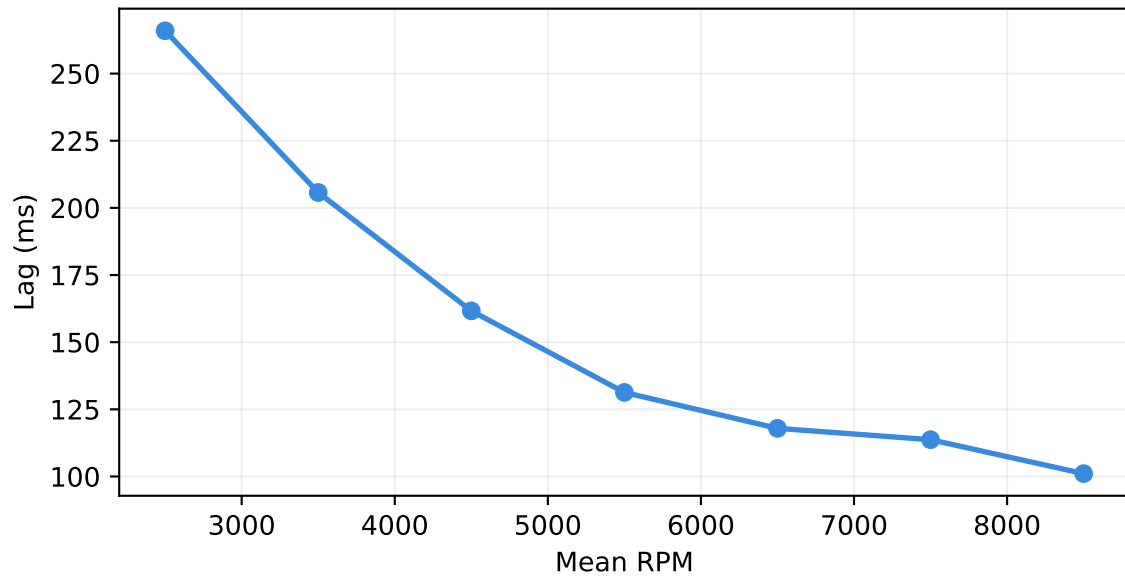


Step 2 — Lag lookup: throttle × throttle-rate

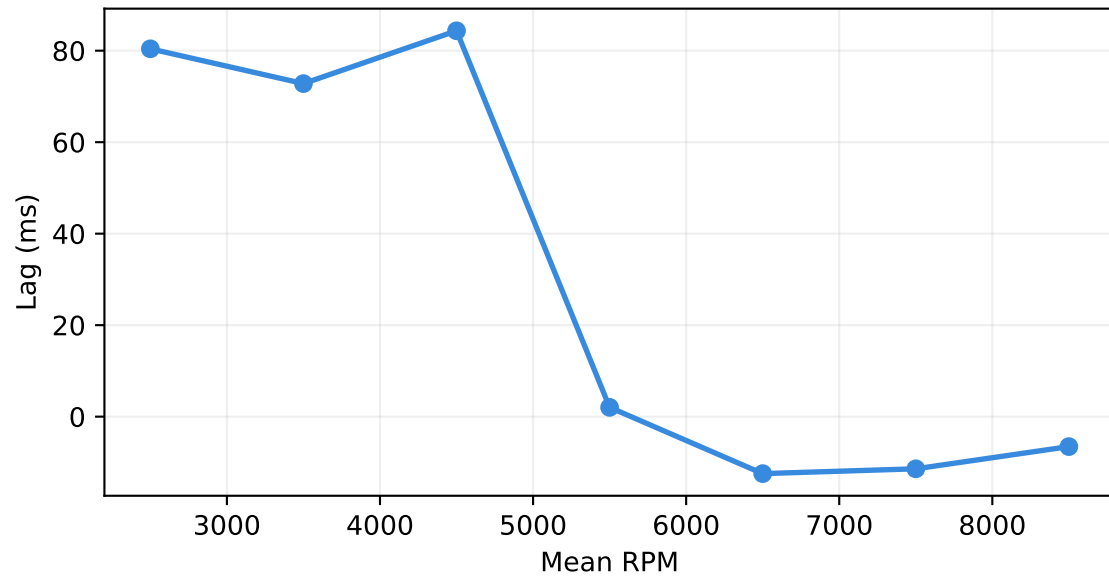


Step 2 — Lag lookup: vs RPM

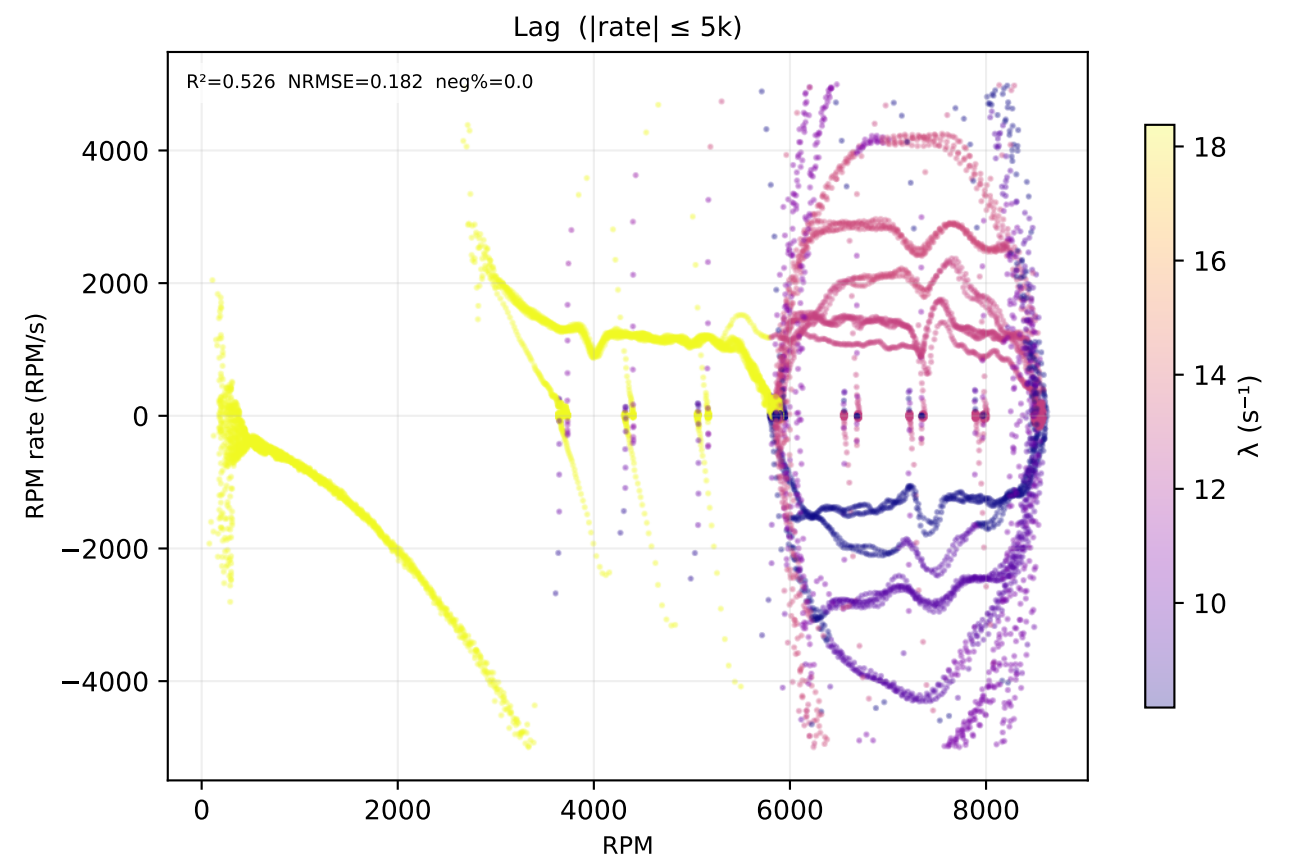
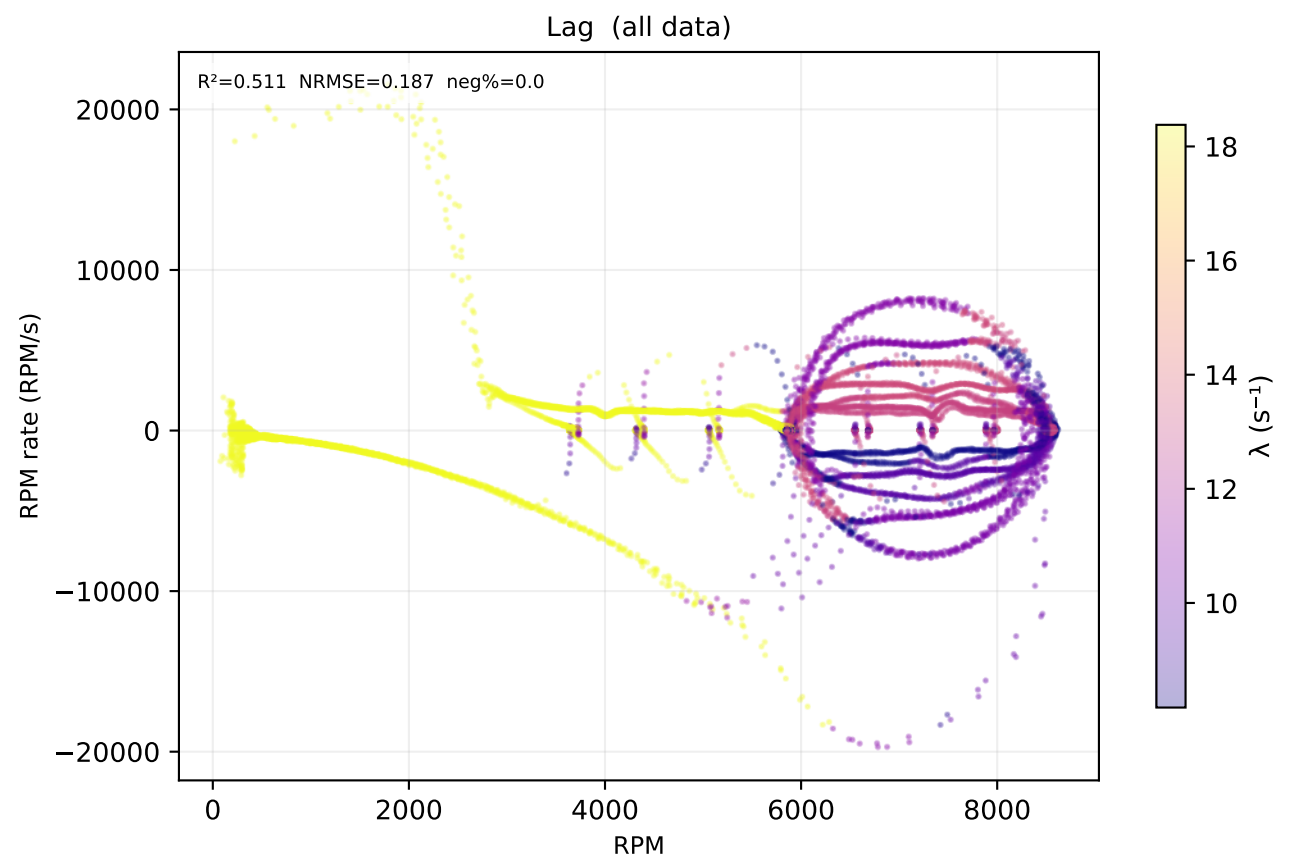
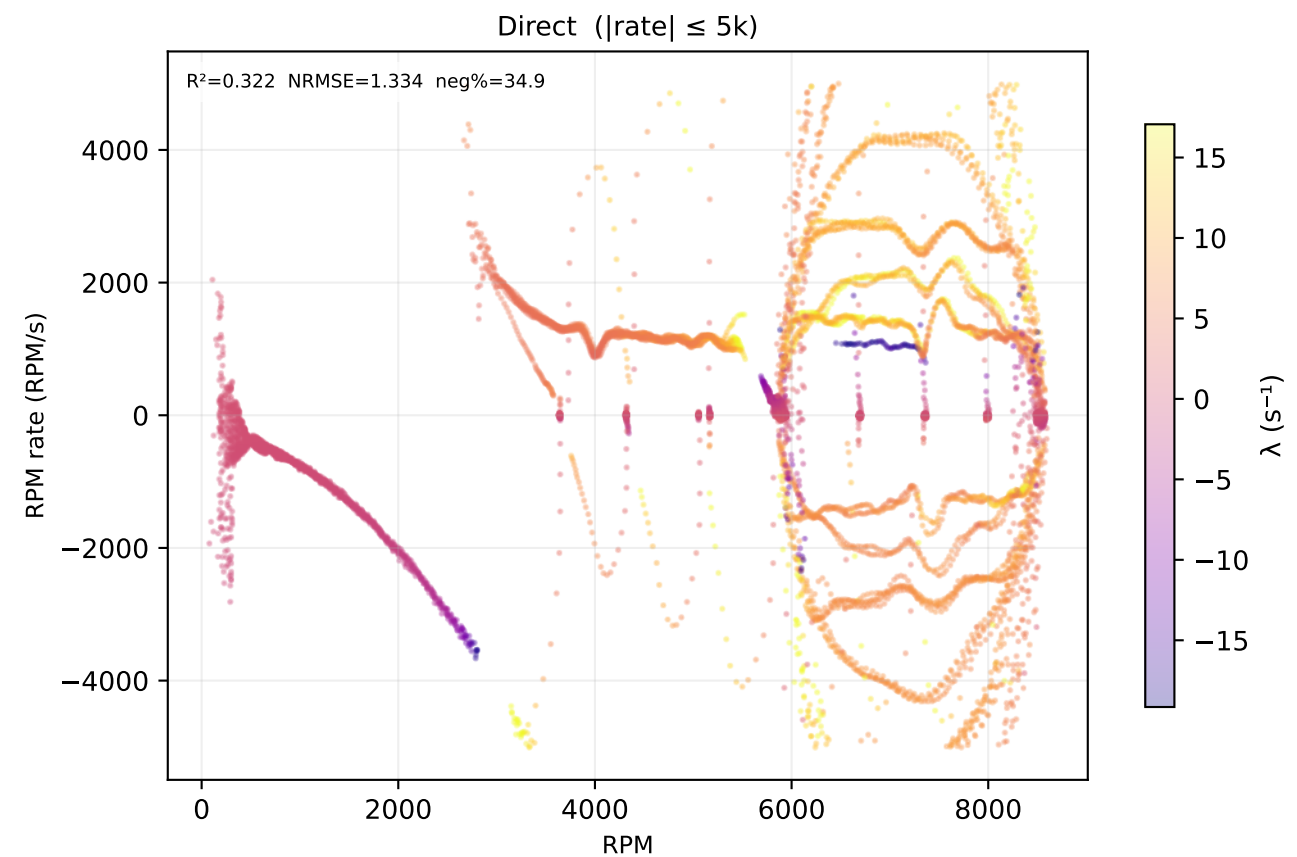
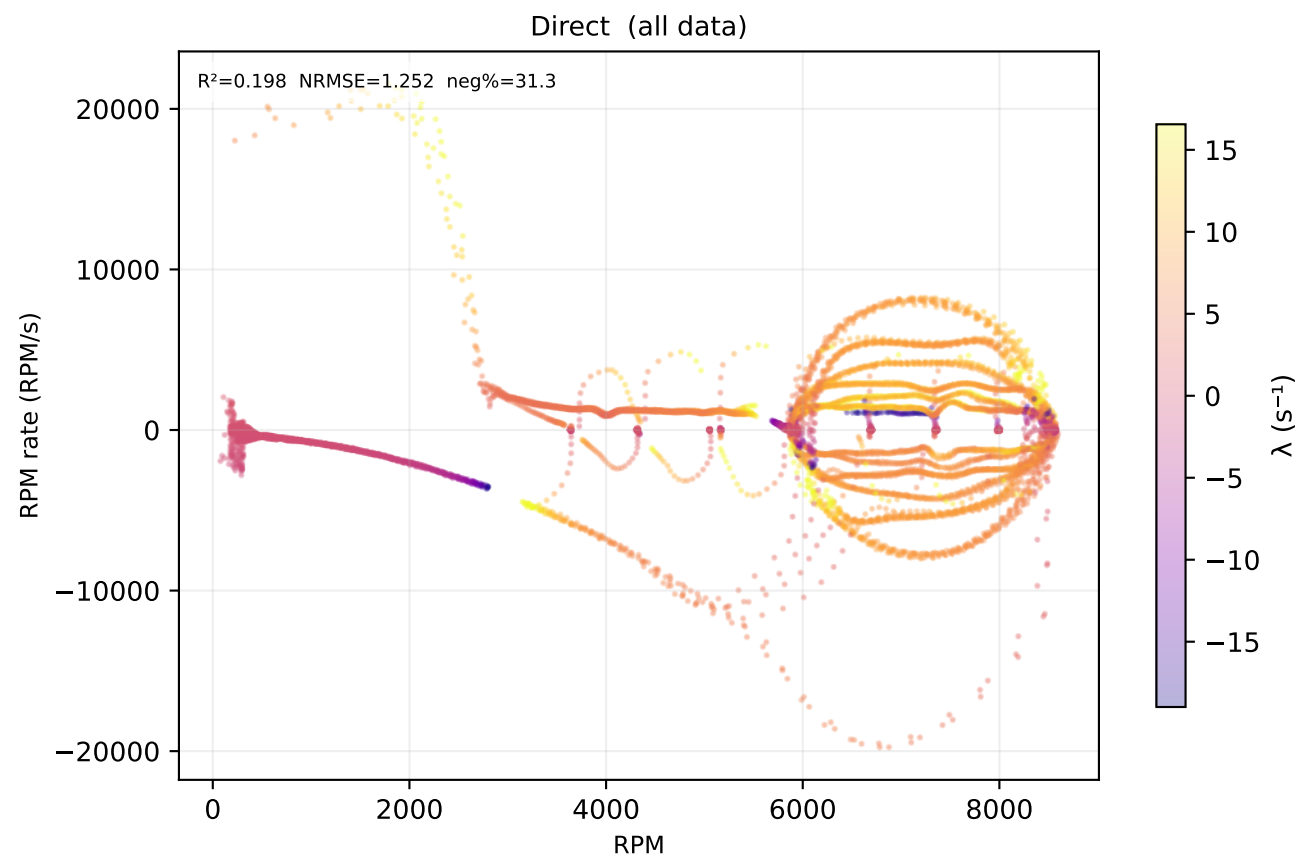
thrust



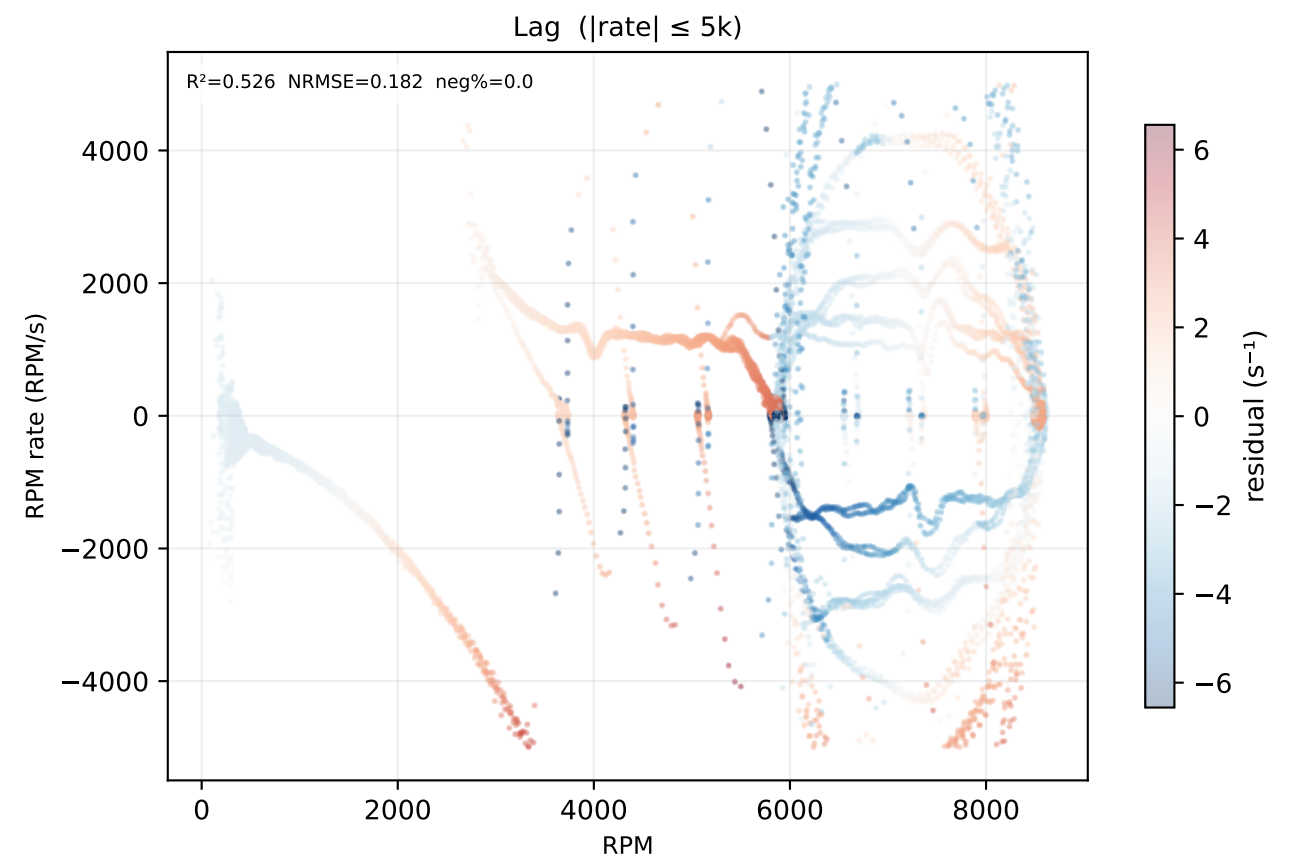
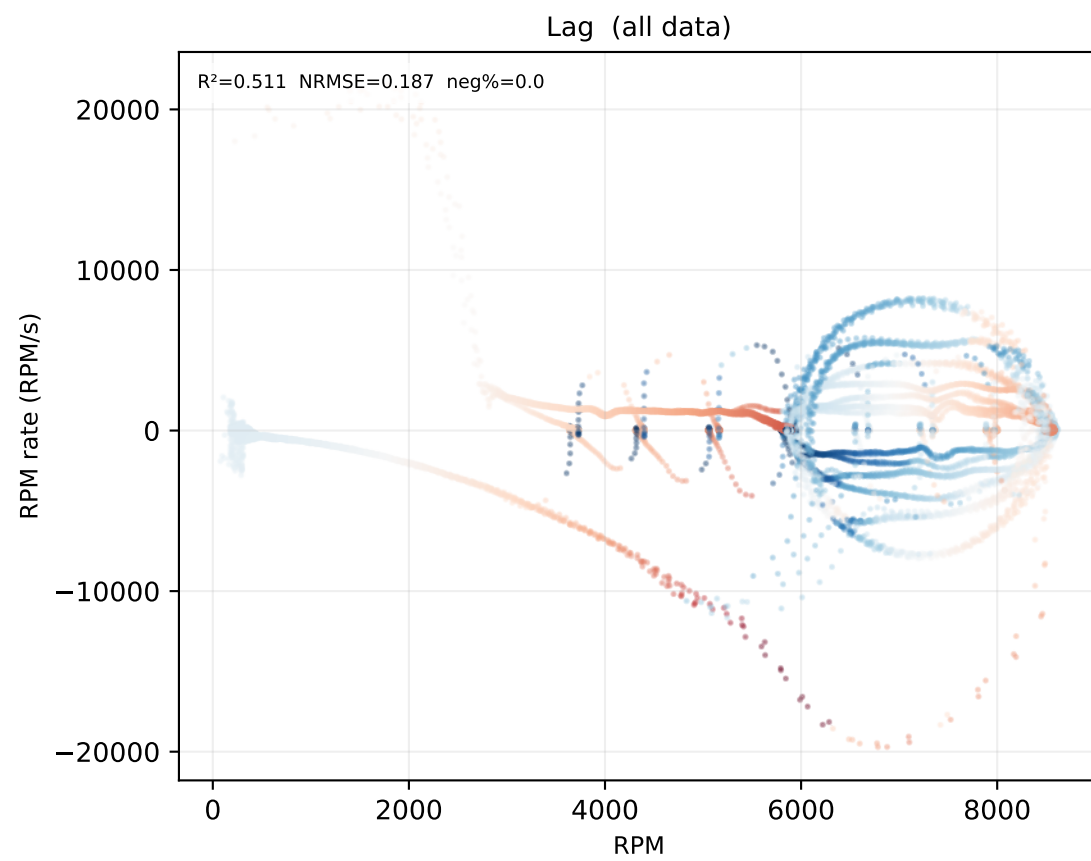
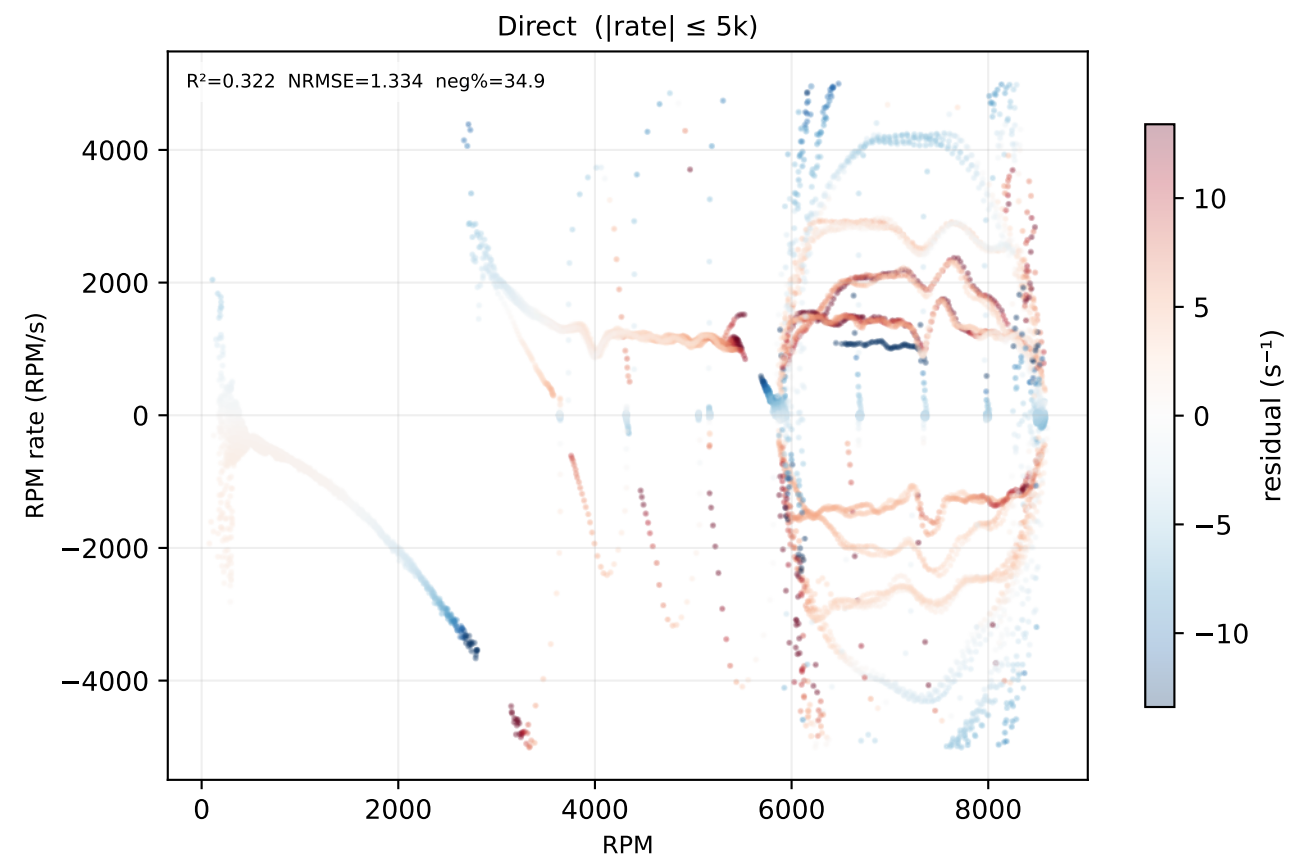
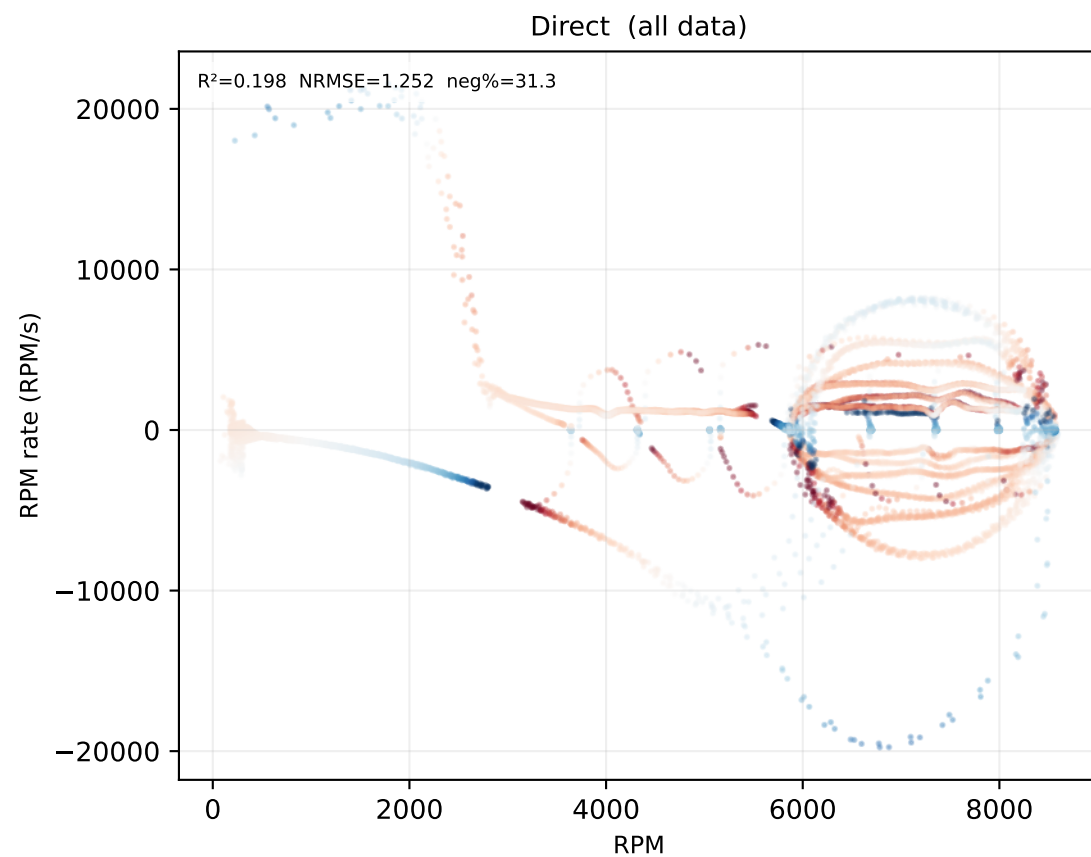
torque



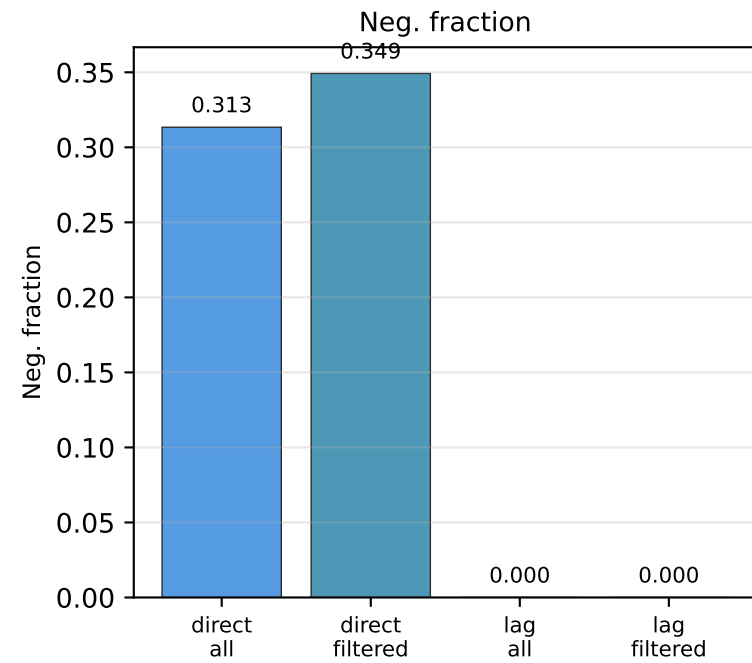
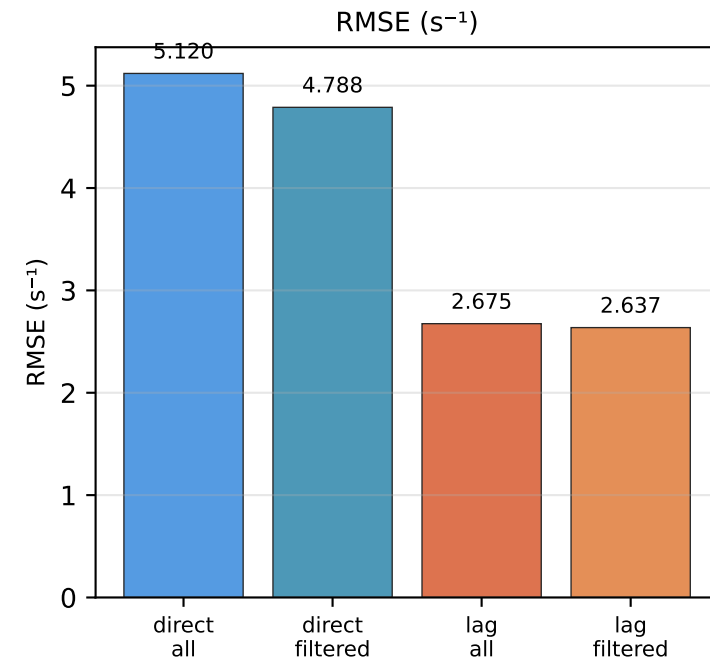
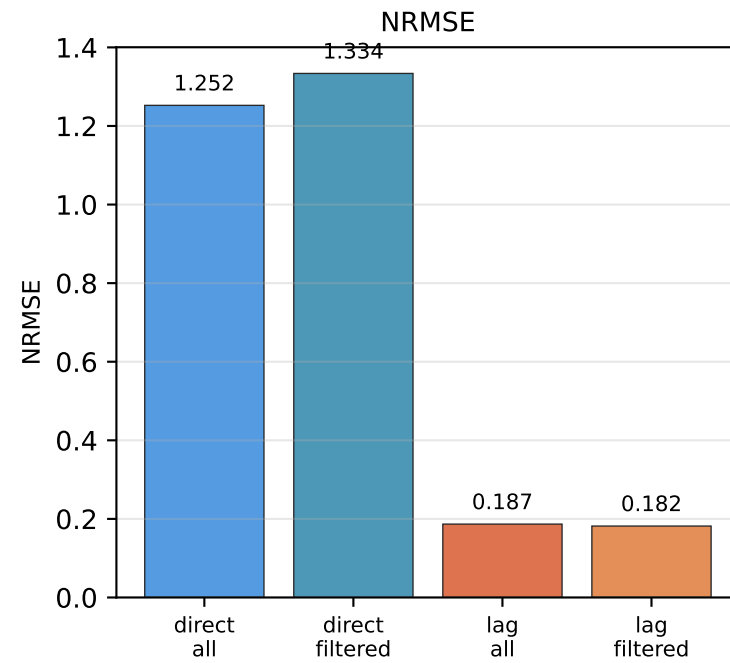
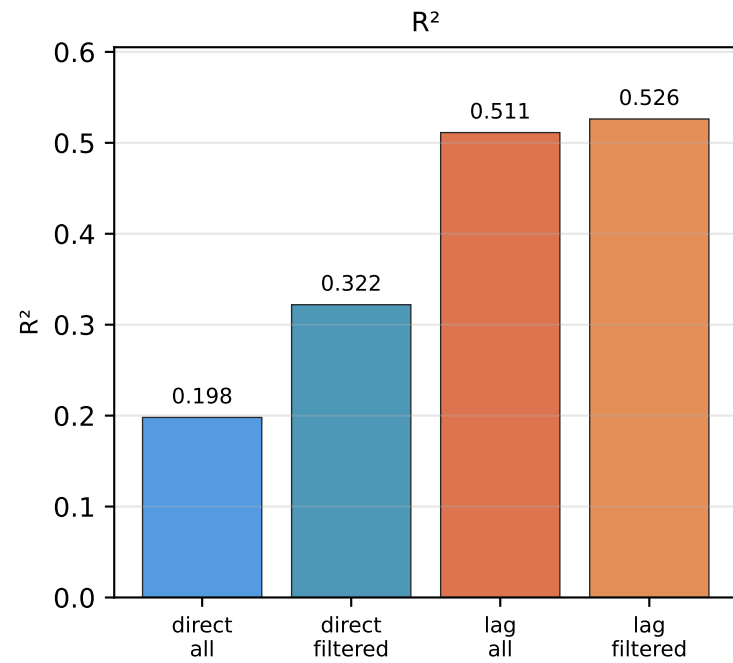
Step 3 — λ values | $\text{RPM_rate} = \lambda \cdot (\text{RPM_cmd} - \text{RPM})$



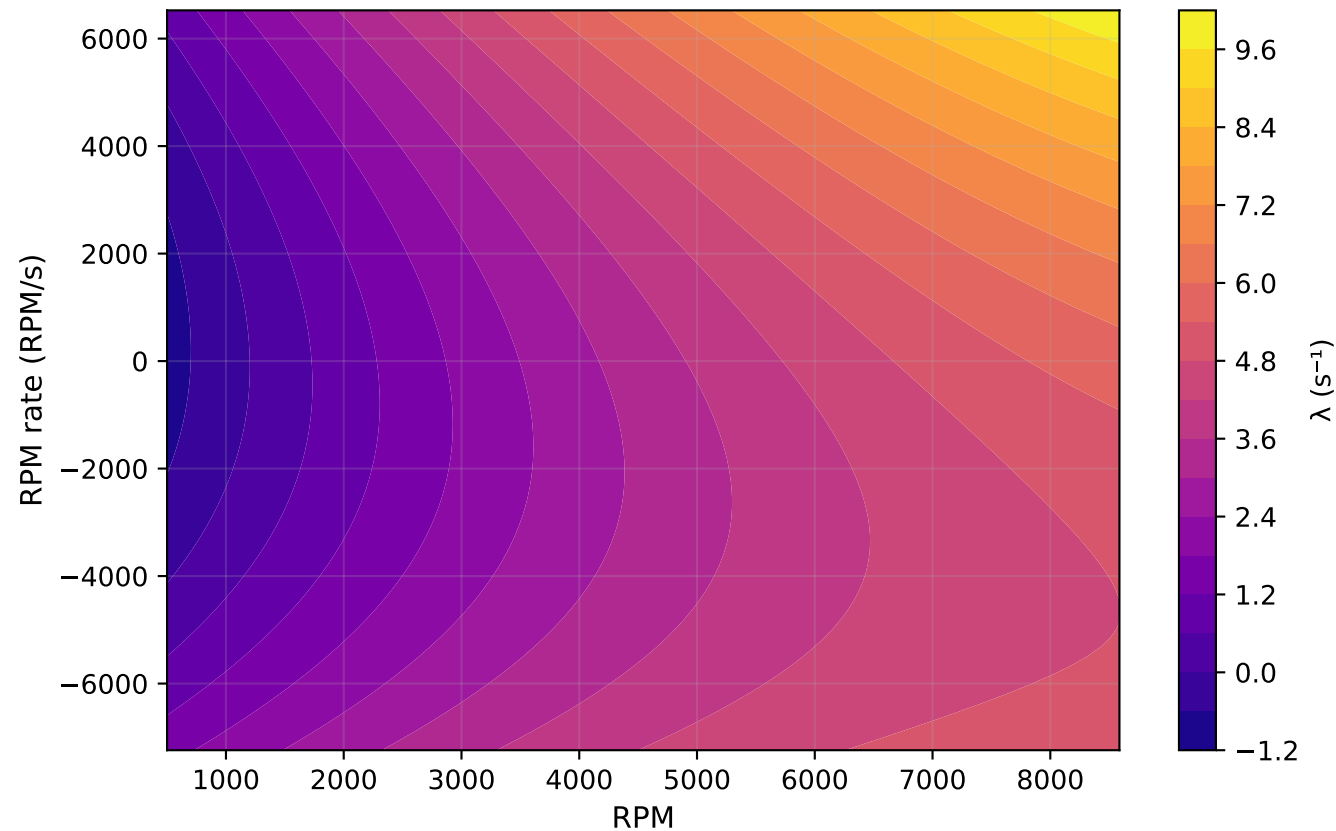
Step 3 — λ residuals (data — fit) | $\text{RPM_rate} = \lambda \cdot (\text{RPM_cmd} - \text{RPM})$



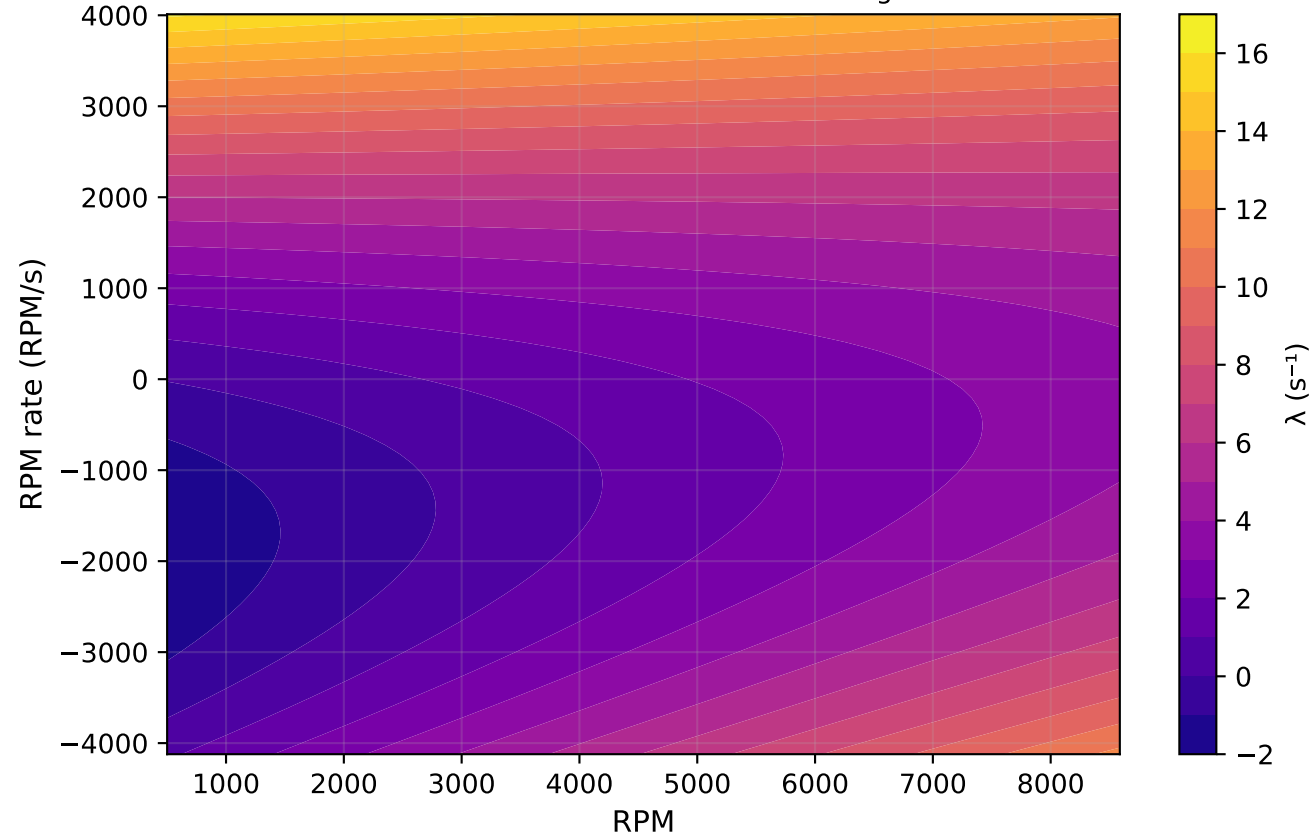
Step 3 — Lambda fit metric comparison



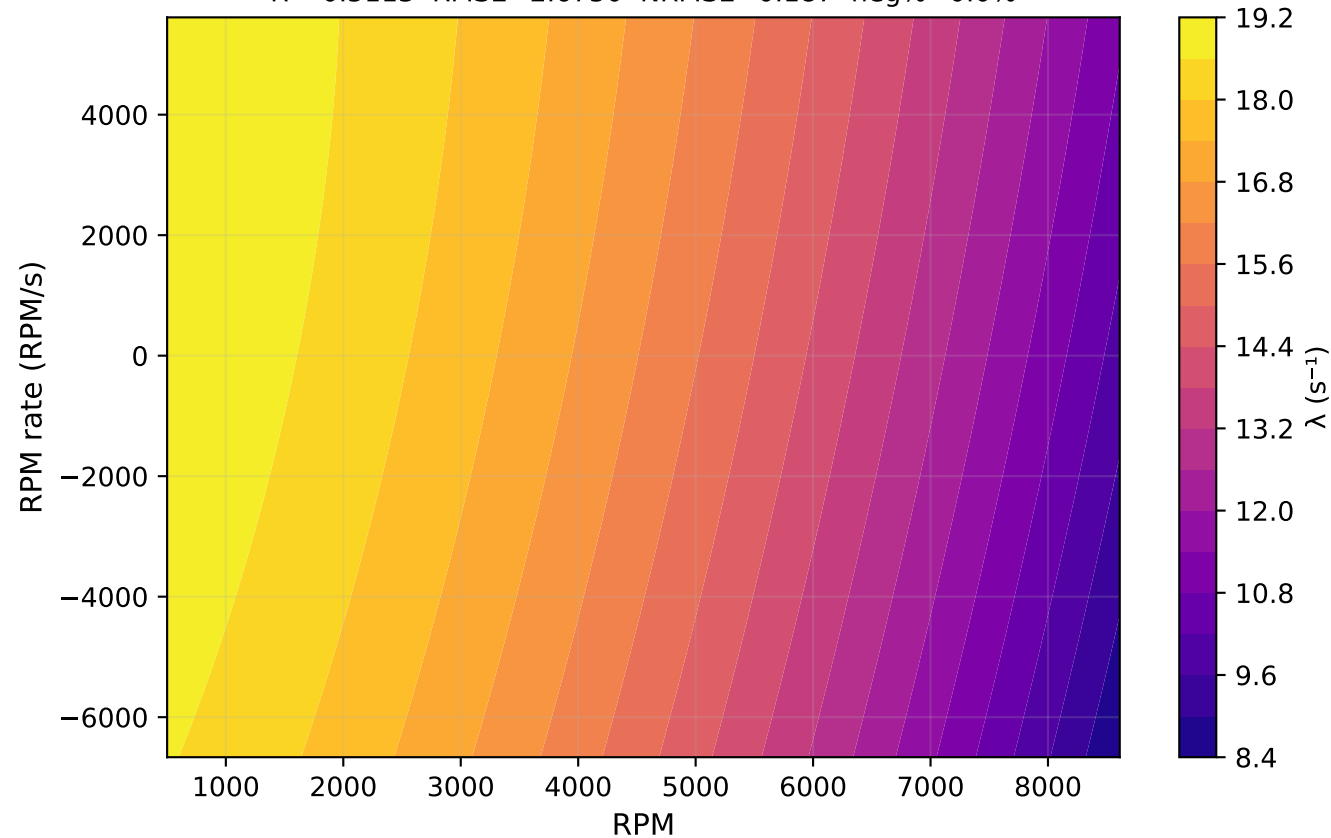
Step 3 — Fitted λ surface: Direct (all data)
 $R^2=0.1980$ RMSE=5.1195 NRMSE=1.252 neg%=31.3%



Step 3 — Fitted λ surface: Direct ($|\text{rate}| \leq 5\text{k}$)
 $R^2=0.3220$ RMSE=4.7884 NRMSE=1.334 neg%=34.9%



Step 3 — Fitted λ surface: Lag (all data)
 $R^2=0.5113$ RMSE=2.6750 NRMSE=0.187 neg%=0.0%



Step 3 — Fitted λ surface: Lag ($|\text{rate}| \leq 5\text{k}$)
 $R^2=0.5262$ RMSE=2.6366 NRMSE=0.182 neg%=0.0%

